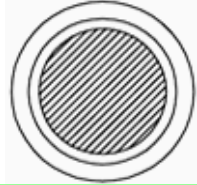


DESCRIPTION	Unit	1C X 400mm ² , A2XY-FRLSH
1 Cable Details		
1.1 kV, Stranded compact circular, Aluminium conductor, XLPE insulated, Overall PVC sheathed - FRLSH cable, Confirming to IS 7098-1.		
2 Material & Dimensional Details		
2.1 Number of Cores	No.	1
2.2 Conductor Cross-Sectional Area	mm ²	400
2.3 Conductor :		
2.3.1 Material		Aluminium conductor
2.3.2 Conductor Cross-Sectional Area (Ph/N)	mm ²	400
2.3.3 Conductor Flexibility class		Class-2
2.3.4 Shape		Stranded compact circular
2.4 Insulation :		
2.4.1 Material		XLPE
2.4.2 Average Thickness	mm	2.00
2.4.3 Minimum Thickness	mm	1.7
2.4.4 Core Identification		As per IS 7098-1
2.5 Inner Sheath :		
2.5.1 Material		Not Applicable
2.5.2 Minimum Thickness	mm	Not applicable
2.6 Armour :		
2.6.1 Material		No Armour
2.6.2 Nominal Size of Armour	mm	Not applicable
2.7 Outer Sheath:		
2.7.1 Material		FRLS
2.7.2 Minimum Thickness	mm	2.20
2.7.3 Colour		Black
2.7.4 Approximate Overall Diameter	mm	32.5 ± 2.0
3 Electrical Parameters		
3.1 Rated Voltage	kV	1.1
3.2 Maximum Conductor DC Resistance at 20°C	ohm/km	0.0778
3.3 Permissible Current Rating in Air at 40°C	A	589
3.4 Maximum Continuous Operating Temperature	°C	90
3.5 Maximum Short Circuit Temperature	°C	250
3.6 Short Circuit Current Rating of Conductor for 1sec.	kA	37.60
4 Packaging		
4.1 Standard Delivery Length	metres	1000
4.2 Individual & overall quantity tolerance	%	± 5
5 Installation Guidelines		
5.1 Minimum Bending Radius	mm	15 x Overall Dia
5.2 Safe Pulling Force	N/mm ²	5 x D ²

FRLS properties of Outer Sheath:

1) Oxygen Index: Min 29. 2) Temp. Index: Min 250 Deg C. 3) Halogen acid gas generation: Max.20% by weight. 4) Smoke density rating: 60% Max.

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Property of Ravin Cables Ltd

DESCRIPTION	Unit	2C X 50mm ² , 2XY-FRLSH
1 Cable Details		
1.1 kV, Stranded sector shaped, Annealed copper conductor, XLPE insulated, Extruded PVC inner sheathed, Overall PVC sheathed - FRLSH cable, Confirming to IS 7098-1.		
2 Material & Dimensional Details		
2.1 Number of Cores	No.	2
2.2 Conductor Cross-Sectional Area	mm ²	50
2.3 Conductor :		
2.3.1 Material		Annealed copper conductor
2.3.2 Conductor Cross-Sectional Area (Ph/N)	mm ²	50
2.3.3 Conductor Flexibility class		Class-2
2.3.4 Shape		Stranded compact sector
2.4 Insulation :		
2.4.1 Material		XLPE
2.4.2 Average Thickness	mm	1.00
2.4.3 Minimum Thickness	mm	0.8
2.4.4 Core Identification		As per IS 7098-1
2.5 Inner Sheath :		
2.5.1 Material		Extruded PVC
2.5.2 Minimum Thickness	mm	0.3
2.6 Armour :		
2.6.1 Material		No Armour
2.6.2 Nominal Size of Armour	mm	Not applicable
2.7 Outer Sheath:		
2.7.1 Material		FRLS
2.7.2 Nominal Thickness	mm	2.00
2.7.3 Colour		Black
2.7.4 Approximate Overall Diameter	mm	21.5 ± 2.0
3 Electrical Parameters		
3.1 Rated Voltage	kV	1.1
3.2 Maximum Conductor DC Resistance at 20°C	ohm/km	0.3870
3.3 Permissible Current Rating in Air at 40°C	A	208
3.4 Maximum Continuous Operating Temperature	°C	90
3.5 Maximum Short Circuit Temperature	°C	250
3.6 Short Circuit Current Rating of Conductor for 1sec.	kA	7.15
4 Packaging		
4.1 Standard Delivery Length	metres	1000
4.2 Individual & overall quantity tolerance	%	± 5
5 Installation Guidelines		
5.1 Minimum Bending Radius	mm	12 x Overall Dia
5.2 Safe Pulling Force with Stocking	N/mm ²	5 x D ²

FRLS properties of Outer Sheath:

1) Oxygen Index: Min 29. 2) Temp. Index: Min 250 Deg C. 3) Halogen acid gas generation: Max.20% by weight. 4) Smoke density rating: 60% Max.

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Property of Ravin Cables Ltd

Customer: Antelec Limited
Offer ref.No.: QL-476
Project: Hansalpur, Gujarat Line - D for Suzuki Motc

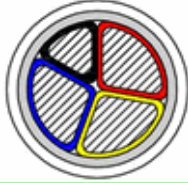
DESCRIPTION	Unit	3.5C X 25mm ² , 2XY-FRLSH
1 Cable Details		
1.1 kV, Stranded sector shaped, Annealed copper conductor, XLPE insulated, Extruded PVC inner sheathed, Overall PVC sheathed - FRLSH cable, Confirming to IS 7098-1.		
2 Material & Dimensional Details		
2.1 Number of Cores	No.	3.5
2.2 Conductor Cross-Sectional Area	mm ²	25
2.3 Conductor : IS 8130		
2.3.1 Material		Annealed copper conductor
2.3.2 Conductor Cross-Sectional Area (Ph/N)	mm ²	25-Ph/16-N
2.3.3 Conductor Flexibility class		Class-2
2.3.4 Shape		Stranded compact sector
2.4 Insulation : IS 7098-1		
2.4.1 Material		XLPE
2.4.2 Average Thickness	mm	0.90 / 0.70
2.4.3 Minimum Thickness	mm	0.71 / 0.53
2.4.4 Core Identification		As per IS 7098-1
2.5 Inner Sheath :		
2.5.1 Material		Extruded PVC
2.5.2 Minimum Thickness	mm	0.3
2.6 Armour : Not Applicable		
2.6.1 Material		No Armour
2.6.2 Nominal Size of Armour	mm	Not applicable
2.7 Outer Sheath: IS 5831		
2.7.1 Material		FRLS
2.7.2 Nominal Thickness	mm	2.00
2.7.3 Colour		Black
2.7.4 Approximate Overall Diameter	mm	22.0 ± 2.0
3 Electrical Parameters		
3.1 Rated Voltage	kV	1.1
3.2 Maximum Conductor DC Resistance at 20°C	ohm/km	0.7270/1.1500
3.3 Permissible Current Rating in Air at 40°C	A	119
3.4 Maximum Continuous Operating Temperature	°C	90
3.5 Maximum Short Circuit Temperature	°C	250
3.6 Short Circuit Current Rating of Conductor for 1sec.	kA	3.58
4 Packaging		
4.1 Standard Delivery Length	metres	1000
4.2 Individual & overall quantity tolerance	%	± 5
5 Installation Guidelines IS 1255		
5.1 Minimum Bending Radius	mm	12 x Overall Dia
5.2 Safe Pulling Force with Stocking	N/mm ²	5 x D ²

FRLS properties of Outer Sheath:

1) Oxygen Index: Min 29. 2) Temp. Index: Min 250 Deg C. 3) Halogen acid gas generation: Max.20% by weight. 4) Smoke density rating: 60% Max.

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Property of Ravin Cables Ltd

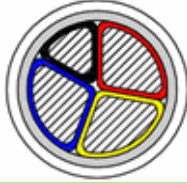
DESCRIPTION	Unit	3.5C X 70mm ² , A2XY-FRLSH
1 Cable Details		
1.1 kV, Stranded sector shaped, Aluminium conductor, XLPE insulated, Extruded PVC inner sheathed, Overall PVC sheathed - FRLSH cable, Confirming to IS 7098-1.		
2 Material & Dimensional Details		
2.1 Number of Cores	No.	3.5
2.2 Conductor Cross-Sectional Area	mm ²	70
2.3 Conductor :		
IS 8130		
2.3.1 Material		Aluminium conductor
2.3.2 Conductor Cross-Sectional Area (Ph/N)	mm ²	70-Ph/35-N
2.3.3 Conductor Flexibility class		Class-2
2.3.4 Shape		Stranded compact sector
2.4 Insulation :		
IS 7098-1		
2.4.1 Material		XLPE
2.4.2 Average Thickness	mm	1.10 / 0.90
2.4.3 Minimum Thickness	mm	0.89 / 0.71
2.4.4 Core Identification		As per IS 7098-1
2.5 Inner Sheath :		
2.5.1 Material		Extruded PVC
2.5.2 Minimum Thickness	mm	0.4
2.6 Armour :		
Not Applicable		
2.6.1 Material		No Armour
2.6.2 Nominal Size of Armour	mm	Not applicable
2.7 Outer Sheath:		
IS 5831		
2.7.1 Material		FRLS
2.7.2 Nominal Thickness	mm	2.20
2.7.3 Colour		Black
2.7.4 Approximate Overall Diameter	mm	30.5 ± 2.0
3 Electrical Parameters		
3.1 Rated Voltage	kV	1.1
3.2 Maximum Conductor DC Resistance at 20°C	ohm/km	0.4430/0.8680
3.3 Permissible Current Rating in Air at 40°C	A	175
3.4 Maximum Continuous Operating Temperature	°C	90
3.5 Maximum Short Circuit Temperature	°C	250
3.6 Short Circuit Current Rating of Conductor for 1sec.	kA	6.58
4 Packaging		
4.1 Standard Delivery Length	metres	1000
4.2 Individual & overall quantity tolerance	%	± 5
5 Installation Guidelines		
IS 1255		
5.1 Minimum Bending Radius	mm	12 x Overall Dia
5.2 Safe Pulling Force with Stocking	N/mm ²	5 x D ²

FRLS properties of Outer Sheath:

1) Oxygen Index: Min 29. 2) Temp. Index: Min 250 Deg C. 3) Halogen acid gas generation: Max.20% by weight. 4) Smoke density rating: 60% Max.

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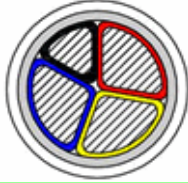
DESCRIPTION	Unit	3.5C X 185mm ² , A2XY-FRLSH
1 Cable Details		
1.1 kV, Stranded sector shaped, Aluminium conductor, XLPE insulated, Extruded PVC inner sheathed, Overall PVC sheathed - FRLSH cable, Confirming to IS 7098-1.		
2 Material & Dimensional Details		
2.1 Number of Cores	No.	3.5
2.2 Conductor Cross-Sectional Area	mm ²	185
2.3 Conductor : IS 8130		
2.3.1 Material		Aluminium conductor
2.3.2 Conductor Cross-Sectional Area (Ph/N)	mm ²	185-Ph/95-N
2.3.3 Conductor Flexibility class		Class-2
2.3.4 Shape		Stranded compact sector
2.4 Insulation : IS 7098-1		
2.4.1 Material		XLPE
2.4.2 Average Thickness	mm	1.60 / 1.10
2.4.3 Minimum Thickness	mm	1.34 / 0.89
2.4.4 Core Identification		As per IS 7098-1
2.5 Inner Sheath :		
2.5.1 Material		Extruded PVC
2.5.2 Minimum Thickness	mm	0.5
2.6 Armour : Not Applicable		
2.6.1 Material		No Armour
2.6.2 Nominal Size of Armour	mm	Not applicable
2.7 Outer Sheath: IS 5831		
2.7.1 Material		FRLS
2.7.2 Nominal Thickness	mm	2.60
2.7.3 Colour		Black
2.7.4 Approximate Overall Diameter	mm	46.0 ± 2.0
3 Electrical Parameters		
3.1 Rated Voltage	kV	1.1
3.2 Maximum Conductor DC Resistance at 20°C	ohm/km	0.1640/0.3200
3.3 Permissible Current Rating in Air at 40°C	A	329
3.4 Maximum Continuous Operating Temperature	°C	90
3.5 Maximum Short Circuit Temperature	°C	250
3.6 Short Circuit Current Rating of Conductor for 1sec.	kA	17.39
4 Packaging		
4.1 Standard Delivery Length	metres	1000
4.2 Individual & overall quantity tolerance	%	± 5
5 Installation Guidelines IS 1255		
5.1 Minimum Bending Radius	mm	12 x Overall Dia
5.2 Safe Pulling Force with Stocking	N/mm ²	5 x D ²

FRLS properties of Outer Sheath:

1) Oxygen Index: Min 29. 2) Temp. Index: Min 250 Deg C. 3) Halogen acid gas generation: Max.20% by weight. 4) Smoke density rating: 60% Max.

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Property of Ravin Cables Ltd

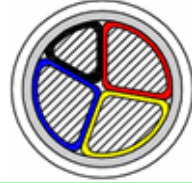
DESCRIPTION	Unit	3.5C X 240mm ² , A2XY-FRLSH
1 Cable Details		
1.1 kV, Stranded sector shaped, Aluminium conductor, XLPE insulated, Extruded PVC inner sheathed, Overall PVC sheathed - FRLSH cable, Confirming to IS 7098-1.		
2 Material & Dimensional Details		
2.1 Number of Cores	No.	3.5
2.2 Conductor Cross-Sectional Area	mm ²	240
2.3 Conductor :		
2.3.1 Material		Aluminium conductor
2.3.2 Conductor Cross-Sectional Area (Ph/N)	mm ²	240-Ph/120-N
2.3.3 Conductor Flexibility class		Class-2
2.3.4 Shape		Stranded compact sector
2.4 Insulation :		
2.4.1 Material		XLPE
2.4.2 Average Thickness	mm	1.70 / 1.20
2.4.3 Minimum Thickness	mm	1.43 / 0.98
2.4.4 Core Identification		As per IS 7098-1
2.5 Inner Sheath :		
2.5.1 Material		Extruded PVC
2.5.2 Minimum Thickness	mm	0.6
2.6 Armour :		
2.6.1 Material		No Armour
2.6.2 Nominal Size of Armour	mm	Not applicable
2.7 Outer Sheath:		
2.7.1 Material		FRLS
2.7.2 Nominal Thickness	mm	2.80
2.7.3 Colour		Black
2.7.4 Approximate Overall Diameter	mm	52.5 ± 3.0
3 Electrical Parameters		
3.1 Rated Voltage	kV	1.1
3.2 Maximum Conductor DC Resistance at 20°C	ohm/km	0.1250/0.2530
3.3 Permissible Current Rating in Air at 40°C	A	392
3.4 Maximum Continuous Operating Temperature	°C	90
3.5 Maximum Short Circuit Temperature	°C	250
3.6 Short Circuit Current Rating of Conductor for 1sec.	kA	22.56
4 Packaging		
4.1 Standard Delivery Length	metres	1000
4.2 Individual & overall quantity tolerance	%	± 5
5 Installation Guidelines		
5.1 Minimum Bending Radius	mm	12 x Overall Dia
5.2 Safe Pulling Force with Stocking	N/mm ²	5 x D ²

FRLS properties of Outer Sheath:

1) Oxygen Index: Min 29. 2) Temp. Index: Min 250 Deg C. 3) Halogen acid gas generation: Max.20% by weight. 4) Smoke density rating: 60% Max.

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Property of Ravin Cables Ltd

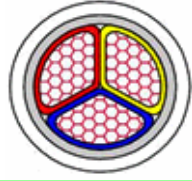
DESCRIPTION	Unit	3.5C X 300mm ² , A2XY-FRLSH
1 Cable Details		
1.1 kV, Stranded sector shaped, Aluminium conductor, XLPE insulated, Extruded PVC inner sheathed, Overall PVC sheathed - FRLSH cable, Confirming to IS 7098-1.		
2 Material & Dimensional Details		
2.1 Number of Cores	No.	3.5
2.2 Conductor Cross-Sectional Area	mm ²	300
2.3 Conductor :		
2.3.1 Material		Aluminium conductor
2.3.2 Conductor Cross-Sectional Area (Ph/N)	mm ²	300-Ph/150-N
2.3.3 Conductor Flexibility class		Class-2
2.3.4 Shape		Stranded compact sector
2.4 Insulation :		
2.4.1 Material		XLPE
2.4.2 Average Thickness	mm	1.80 / 1.40
2.4.3 Minimum Thickness	mm	1.52 / 1.16
2.4.4 Core Identification		As per IS 7098-1
2.5 Inner Sheath :		
2.5.1 Material		Extruded PVC
2.5.2 Minimum Thickness	mm	0.6
2.6 Armour :		
2.6.1 Material		No Armour
2.6.2 Nominal Size of Armour	mm	Not applicable
2.7 Outer Sheath:		
2.7.1 Material		FRLS
2.7.2 Nominal Thickness	mm	3.00
2.7.3 Colour		Black
2.7.4 Approximate Overall Diameter	mm	57.5 ± 3.0
3 Electrical Parameters		
3.1 Rated Voltage	kV	1.1
3.2 Maximum Conductor DC Resistance at 20°C	ohm/km	0.1000/0.2060
3.3 Permissible Current Rating in Air at 40°C	A	452
3.4 Maximum Continuous Operating Temperature	°C	90
3.5 Maximum Short Circuit Temperature	°C	250
3.6 Short Circuit Current Rating of Conductor for 1sec.	kA	28.20
4 Packaging		
4.1 Standard Delivery Length	metres	500
4.2 Individual & overall quantity tolerance	%	± 5
5 Installation Guidelines		
5.1 Minimum Bending Radius	mm	12 x Overall Dia
5.2 Safe Pulling Force with Stocking	N/mm ²	5 x D ²

FRLS properties of Outer Sheath:

1) Oxygen Index: Min 29. 2) Temp. Index: Min 250 Deg C. 3) Halogen acid gas generation: Max.20% by weight. 4) Smoke density rating: 60% Max.

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Property of Ravin Cables Ltd

DESCRIPTION	Unit	3C X 50mm ² , 2XY-FRLSH
1 Cable Details		
1.1 kV, Stranded sector shaped, Annealed copper conductor, XLPE insulated, Extruded PVC inner sheathed, Overall PVC sheathed - FRLSH cable, Confirming to IS 7098-1.		
2 Material & Dimensional Details		
2.1 Number of Cores	No.	3
2.2 Conductor Cross-Sectional Area	mm ²	50
2.3 Conductor :		
2.3.1 Material		Annealed copper conductor
2.3.2 Conductor Cross-Sectional Area (Ph/N)	mm ²	50
2.3.3 Conductor Flexibility class		Class-2
2.3.4 Shape		Stranded compact sector
2.4 Insulation :		
2.4.1 Material		XLPE
2.4.2 Average Thickness	mm	1.00
2.4.3 Minimum Thickness	mm	0.8
2.4.4 Core Identification		As per IS 7098-1
2.5 Inner Sheath :		
2.5.1 Material		Extruded PVC
2.5.2 Minimum Thickness	mm	0.3
2.6 Armour :		
2.6.1 Material		No Armour
2.6.2 Nominal Size of Armour	mm	Not applicable
2.7 Outer Sheath:		
2.7.1 Material		FRLS
2.7.2 Nominal Thickness	mm	2.00
2.7.3 Colour		Black
2.7.4 Approximate Overall Diameter	mm	24.5 ± 2.0
3 Electrical Parameters		
3.1 Rated Voltage	kV	1.1
3.2 Maximum Conductor DC Resistance at 20°C	ohm/km	0.3870
3.3 Permissible Current Rating in Air at 40°C	A	179
3.4 Maximum Continuous Operating Temperature	°C	90
3.5 Maximum Short Circuit Temperature	°C	250
3.6 Short Circuit Current Rating of Conductor for 1sec.	kA	7.15
4 Packaging		
4.1 Standard Delivery Length	metres	1000
4.2 Individual & overall quantity tolerance	%	± 5
5 Installation Guidelines		
5.1 Minimum Bending Radius	mm	12 x Overall Dia
5.2 Safe Pulling Force with Stocking	N/mm ²	5 x D ²

FRLS properties of Outer Sheath:

1) Oxygen Index: Min 29. 2) Temp. Index: Min 250 Deg C. 3) Halogen acid gas generation: Max.20% by weight. 4) Smoke density rating: 60% Max.

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Property of Ravin Cables Ltd

Customer: Antelec Limited
Offer ref.No.: QL-476
Project: Hansalpur, Gujarat Line - D for Suzuki Motc

DESCRIPTION	Unit	3C X 70mm ² , A2XIFY-FRLSH
1 Cable Details		
11/11 KV (UE), Stranded compact circular, Aluminium conductor, Conductor screen, XLPE insulation, Insulation screen, Copper Tape Screened, Extruded PVC inner sheathed, Galvanised steel flat strip armour, Overall PVC sheathed - FRLSH cable, Confirming to IS 7098-2.		
2 Material & Dimensional Details		
2.1 Number of Cores	No.	3
2.2 Conductor Cross-Sectional Area	mm ²	70
2.3 Conductor :		
2.3.1 Material		Aluminium conductor
2.3.2 Conductor Cross-Sectional Area (Ph/N)	mm ²	70
2.3.3 Conductor Flexibility class		Class-2
2.3.4 Shape		Stranded compact circular
2.3.5 Approximate Diameter of Conductor	mm	10.0 ± 0.20
2.4 Conductor Screen :		
2.4.1 Material		Extruded Semi-Con Material
2.4.2 Non-Metallic - Approximate thickness	mm	0.30
2.5 Insulation :		
2.5.1 Material		HT-XLPE
2.5.2 Average Thickness	mm	5.5
2.5.3 Minimum Thickness	mm	4.85
2.6 Insulation Screen :		
2.6.1 Material		Extruded Semi-Con Material
2.6.2 Non-Metallic - Approximate thickness	mm	0.30
2.7 Mettalic Screen :		
2.7.1 Material		Individual Copper Tape
2.7.2 Metallic -Nominal Thickness	mm	0.03
2.7.3 Core Identification		As per IS 7098-2
2.8 Inner Sheath :		
2.8.1 Material		Extruded PVC
2.8.2 Minimum Thickness	mm	0.6
2.9 Armour :		
2.9.1 Material		Galvanised steel flat Strip
2.9.2 Nominal Size of Armour	mm	4.0 X 0.8
2.10 Outer Sheath :		
2.10.1 Material		FRLS
2.10.2 Minimum Thickness	mm	2.20
2.10.3 Colour		Black
2.10.4 Approximate Overall Diameter	mm	57.0 ± 3.0
3 Electrical Parameters		
3.1 Rated Voltage	kV	11/11 KV (UE)
3.2 Maximum Conductor DC Resistance at 20°C	ohm/km	0.4430
3.3 Permissible Current Rating in Air at 40°C	A	184
3.4 Maximum Continous Operating Temperature	°C	90
3.5 Maximum Short Circuit Temperature	°C	250
3.6 Short Circuit Current Rating of Conductor for 1sec.	kA	6.58
4 Packaging		
4.1 Standard Delivery Length	metres	1000
4.2 Individual & overall quantity tolerance	%	± 5
5 Installation Guidelines		
5.1 Minimum Bending Radius	mm	15 x Overall Dia
5.2 Safe Pulling Force with Pulling eye	N/mm ²	30

FRLS properties of Outer Sheath:

1) Oxygen Index: Min 29. 2) Temp. Index: Min 250 Deg C. 3) Halogen acid gas generation: Max.20% by weight. 4) Smoke density rating: 60% Max.

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Property of Ravin Cables Ltd

Customer: Antelec Limited
Offer ref.No.: QL-476
Project: Hansalpur, Gujarat Line - D for Suzuki Motc

DESCRIPTION	Unit	3C X 185mm ² , A2XIFY-FRLSH
1 Cable Details		
11/11 KV (UE), Stranded compact circular, Aluminium conductor, Conductor screen, XLPE insulation, Insulation screen, Copper Tape Screened, Extruded PVC inner sheathed, Galvanised steel flat strip armour, Overall PVC sheathed - FRLSH cable, Confirming to IS 7098-2.		
2 Material & Dimensional Details		
2.1 Number of Cores	No.	3
2.2 Conductor Cross-Sectional Area	mm ²	185
2.3 Conductor :		
2.3.1 Material		Aluminium conductor
2.3.2 Conductor Cross-Sectional Area (Ph/N)	mm ²	185
2.3.3 Conductor Flexibility class		Class-2
2.3.4 Shape		Stranded compact circular
2.3.5 Approximate Diameter of Conductor	mm	16.0 ± 0.20
2.4 Conductor Screen :		
2.4.1 Material		Extruded Semi-Con Material
2.4.2 Non-Metallic - Approximate thickness	mm	0.30
2.5 Insulation :		
2.5.1 Material		HT-XLPE
2.5.2 Average Thickness	mm	5.5
2.5.3 Minimum Thickness	mm	4.85
2.6 Insulation Screen :		
2.6.1 Material		Extruded Semi-Con Material
2.6.2 Non-Metallic - Approximate thickness	mm	0.30
2.7 Mettalic Screen :		
2.7.1 Material		Individual Copper Tape
2.7.2 Metallic -Nominal Thickness	mm	0.03
2.7.3 Core Identification		As per IS 7098-2
2.8 Inner Sheath :		
2.8.1 Material		Extruded PVC
2.8.2 Minimum Thickness	mm	0.7
2.9 Armour :		
2.9.1 Material		Galvanised steel flat Strip
2.9.2 Nominal Size of Armour	mm	4.0 X 0.8
2.10 Outer Sheath :		
2.10.1 Material		FRLS
2.10.2 Minimum Thickness	mm	2.68
2.10.3 Colour		Black
2.10.4 Approximate Overall Diameter	mm	73.5 ± 3.0
3 Electrical Parameters		
3.1 Rated Voltage	kV	11/11 KV (UE)
3.2 Maximum Conductor DC Resistance at 20°C	ohm/km	0.1640
3.3 Permissible Current Rating in Air at 40°C	A	330
3.4 Maximum Continous Operating Temperature	°C	90
3.5 Maximum Short Circuit Temperature	°C	250
3.6 Short Circuit Current Rating of Conductor for 1sec.	kA	17.39
4 Packaging		
4.1 Standard Delivery Length	metres	500
4.2 Individual & overall quantity tolerance	%	± 5
5 Installation Guidelines		
5.1 Minimum Bending Radius	mm	15 x Overall Dia
5.2 Safe Pulling Force with Pulling eye	N/mm ²	30

FRLS properties of Outer Sheath:

1) Oxygen Index: Min 29. 2) Temp. Index: Min 250 Deg C. 3) Halogen acid gas generation: Max.20% by weight. 4) Smoke density rating: 60% Max.

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Property of Ravin Cables Ltd

Customer: Antelec Limited
Offer ref.No.: QL-476
Project: Hansalpur, Gujarat Line - D for Suzuki Motc

DESCRIPTION	Unit	3C X 300mm ² , A2XIFY-FRLSH
1 Cable Details		
11/11 KV (UE), Stranded compact circular, Aluminium conductor, Conductor screen, XLPE insulation, Insulation screen, Copper Tape Screened, Extruded PVC inner sheathed, Galvanised steel flat strip armour, Overall PVC sheathed - FRLSH cable, Confirming to IS 7098-2.		
2 Material & Dimensional Details		
2.1 Number of Cores	No.	3
2.2 Conductor Cross-Sectional Area	mm ²	300
2.3 Conductor :		
2.3.1 Material		Aluminium conductor
2.3.2 Conductor Cross-Sectional Area (Ph/N)	mm ²	300
2.3.3 Conductor Flexibility class		Class-2
2.3.4 Shape		Stranded compact circular
2.3.5 Approximate Diameter of Conductor	mm	20.2 ± 0.20
2.4 Conductor Screen :		
2.4.1 Material		Extruded Semi-Con Material
2.4.2 Non-Metallic - Approximate thickness	mm	0.30
2.5 Insulation :		
2.5.1 Material		HT-XLPE
2.5.2 Average Thickness	mm	5.5
2.5.3 Minimum Thickness	mm	4.85
2.6 Insulation Screen :		
2.6.1 Material		Extruded Semi-Con Material
2.6.2 Non-Metallic - Approximate thickness	mm	0.30
2.7 Mettalic Screen :		
2.7.1 Material		Individual Copper Tape
2.7.2 Metallic -Nominal Thickness	mm	0.03
2.7.3 Core Identification		As per IS 7098-2
2.8 Inner Sheath :		
2.8.1 Material		Extruded PVC
2.8.2 Minimum Thickness	mm	0.7
2.9 Armour :		
2.9.1 Material		Galvanised steel flat Strip
2.9.2 Nominal Size of Armour	mm	4.0 X 0.8
2.10 Outer Sheath :		
2.10.1 Material		FRLS
2.10.2 Minimum Thickness	mm	3.00
2.10.3 Colour		Black
2.10.4 Approximate Overall Diameter	mm	83.0 ± 3.0
3 Electrical Parameters		
3.1 Rated Voltage	kV	11/11 KV (UE)
3.2 Maximum Conductor DC Resistance at 20°C	ohm/km	0.1000
3.3 Permissible Current Rating in Air at 40°C	A	441
3.4 Maximum Continous Operating Temperature	°C	90
3.5 Maximum Short Circuit Temperature	°C	250
3.6 Short Circuit Current Rating of Conductor for 1sec.	kA	28.20
4 Packaging		
4.1 Standard Delivery Length	metres	500
4.2 Individual & overall quantity tolerance	%	± 5
5 Installation Guidelines		
5.1 Minimum Bending Radius	mm	15 x Overall Dia
5.2 Safe Pulling Force with Pulling eye	N/mm ²	30

FRLS properties of Outer Sheath:

1) Oxygen Index: Min 29. 2) Temp. Index: Min 250 Deg C. 3) Halogen acid gas generation: Max.20% by weight. 4) Smoke density rating: 60% Max.

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Property of Ravin Cables Ltd

Customer: Antelec Limited
Offer ref.No.: QL-476
Project: Hansalpur, Gujarat Line - D for Suzuki Motc

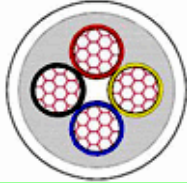
DESCRIPTION	Unit	3C X 400mm ² , A2XIFY-FRLSH
1 Cable Details		
11/11 KV (UE), Stranded compact circular, Aluminium conductor, Conductor screen, XLPE insulation, Insulation screen, Copper Tape Screened, Extruded PVC inner sheathed, Galvanised steel flat strip armour, Overall PVC sheathed - FRLSH cable, Confirming to IS 7098-2.		
2 Material & Dimensional Details		
2.1 Number of Cores	No.	3
2.2 Conductor Cross-Sectional Area	mm ²	400
2.3 Conductor :		
2.3.1 Material		Aluminium conductor
2.3.2 Conductor Cross-Sectional Area (Ph/N)	mm ²	400
2.3.3 Conductor Flexibility class		Class-2
2.3.4 Shape		Stranded compact circular
2.3.5 Approximate Diameter of Conductor	mm	23.2 ± 0.20
2.4 Conductor Screen :		
2.4.1 Material		Extruded Semi-Con Material
2.4.2 Non-Metallic - Approximate thickness	mm	0.30
2.5 Insulation :		
2.5.1 Material		HT-XLPE
2.5.2 Average Thickness	mm	5.5
2.5.3 Minimum Thickness	mm	4.85
2.6 Insulation Screen :		
2.6.1 Material		Extruded Semi-Con Material
2.6.2 Non-Metallic - Approximate thickness	mm	0.30
2.7 Mettalic Screen :		
2.7.1 Material		Individual Copper Tape
2.7.2 Metallic -Nominal Thickness	mm	0.03
2.7.3 Core Identification		As per IS 7098-2
2.8 Inner Sheath :		
2.8.1 Material		Extruded PVC
2.8.2 Minimum Thickness	mm	0.7
2.9 Armour :		
2.9.1 Material		Galvanised steel flat Strip
2.9.2 Nominal Size of Armour	mm	4.0 X 0.8
2.10 Outer Sheath :		
2.10.1 Material		FRLS
2.10.2 Minimum Thickness	mm	3.00
2.10.3 Colour		Black
2.10.4 Approximate Overall Diameter	mm	89.5 ± 3.0
3 Electrical Parameters		
3.1 Rated Voltage	kV	11/11 KV (UE)
3.2 Maximum Conductor DC Resistance at 20°C	ohm/km	0.0778
3.3 Permissible Current Rating in Air at 40°C	A	512
3.4 Maximum Continous Operating Temperature	°C	90
3.5 Maximum Short Circuit Temperature	°C	250
3.6 Short Circuit Current Rating of Conductor for 1sec.	kA	37.60
4 Packaging		
4.1 Standard Delivery Length	metres	250
4.2 Individual & overall quantity tolerance	%	± 5
5 Installation Guidelines		
5.1 Minimum Bending Radius	mm	15 x Overall Dia
5.2 Safe Pulling Force with Pulling eye	N/mm ²	30

FRLS properties of Outer Sheath:

1) Oxygen Index: Min 29. 2) Temp. Index: Min 250 Deg C. 3) Halogen acid gas generation: Max.20% by weight. 4) Smoke density rating: 60% Max.

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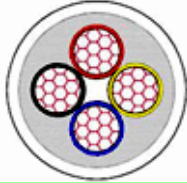
DESCRIPTION	Unit	4C X 2.5mm ² , 2XY-FRLSH
1 Cable Details		
1.1 kV, Stranded circular, Annealed copper conductor, XLPE insulated, Extruded PVC inner sheathed, Overall PVC sheathed - FRLSH cable, Confirming to IS 7098-1.		
2 Material & Dimensional Details		
2.1 Number of Cores	No.	4
2.2 Conductor Cross-Sectional Area	mm ²	2.5
2.3 Conductor : IS 8130		
2.3.1 Material		Annealed copper conductor
2.3.2 Conductor Cross-Sectional Area (Ph/N)	mm ²	2.5
2.3.3 Conductor Flexibility class		Class-2
2.3.4 Shape		Stranded circular
2.4 Insulation : IS 7098-1		
2.4.1 Material		XLPE
2.4.2 Average Thickness	mm	0.70
2.4.3 Minimum Thickness	mm	0.53
2.4.4 Core Identification		As per IS 7098-1
2.5 Inner Sheath :		
2.5.1 Material		Extruded PVC
2.5.2 Minimum Thickness	mm	0.3
2.6 Armour : Not Applicable		
2.6.1 Material		No Armour
2.6.2 Nominal Size of Armour	mm	Not applicable
2.7 Outer Sheath: IS 5831		
2.7.1 Material		FRLS
2.7.2 Nominal Thickness	mm	1.80
2.7.3 Colour		Black
2.7.4 Approximate Overall Diameter	mm	14.0 ± 2.0
3 Electrical Parameters		
3.1 Rated Voltage	kV	1.1
3.2 Maximum Conductor DC Resistance at 20°C	ohm/km	7.4100
3.3 Permissible Current Rating in Air at 40°C	A	30
3.4 Maximum Continuous Operating Temperature	°C	90
3.5 Maximum Short Circuit Temperature	°C	250
3.6 Short Circuit Current Rating of Conductor for 1sec.	kA	0.36
4 Packaging		
4.1 Standard Delivery Length	metres	1000
4.2 Individual & overall quantity tolerance	%	± 5
5 Installation Guidelines IS 1255		
5.1 Minimum Bending Radius	mm	12 x Overall Dia
5.2 Safe Pulling Force with Stocking	N/mm ²	5 x D ²

FRLS properties of Outer Sheath:

1) Oxygen Index: Min 29. 2) Temp. Index: Min 250 Deg C. 3) Halogen acid gas generation: Max.20% by weight. 4) Smoke density rating: 60% Max.

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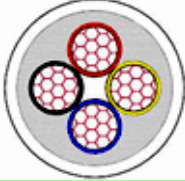
DESCRIPTION	Unit	4C X 4mm ² , 2XY-FRLSH
1 Cable Details		
1.1 kV, Stranded circular, Annealed copper conductor, XLPE insulated, Extruded PVC inner sheathed, Overall PVC sheathed - FRLSH cable, Confirming to IS 7098-1.		
2 Material & Dimensional Details		
2.1 Number of Cores	No.	4
2.2 Conductor Cross-Sectional Area	mm ²	4
2.3 Conductor :		
2.3.1 Material		Annealed copper conductor
2.3.2 Conductor Cross-Sectional Area (Ph/N)	mm ²	4
2.3.3 Conductor Flexibility class		Class-2
2.3.4 Shape		Stranded circular
2.4 Insulation :		
2.4.1 Material		XLPE
2.4.2 Average Thickness	mm	0.70
2.4.3 Minimum Thickness	mm	0.53
2.4.4 Core Identification		As per IS 7098-1
2.5 Inner Sheath :		
2.5.1 Material		Extruded PVC
2.5.2 Minimum Thickness	mm	0.3
2.6 Armour :		
2.6.1 Material		No Armour
2.6.2 Nominal Size of Armour	mm	Not applicable
2.7 Outer Sheath:		
2.7.1 Material		FRLS
2.7.2 Nominal Thickness	mm	1.80
2.7.3 Colour		Black
2.7.4 Approximate Overall Diameter	mm	15.0 ± 2.0
3 Electrical Parameters		
3.1 Rated Voltage	kV	1.1
3.2 Maximum Conductor DC Resistance at 20°C	ohm/km	4.6100
3.3 Permissible Current Rating in Air at 40°C	A	41
3.4 Maximum Continuous Operating Temperature	°C	90
3.5 Maximum Short Circuit Temperature	°C	250
3.6 Short Circuit Current Rating of Conductor for 1sec.	kA	0.57
4 Packaging		
4.1 Standard Delivery Length	metres	1000
4.2 Individual & overall quantity tolerance	%	± 5
5 Installation Guidelines		
5.1 Minimum Bending Radius	mm	12 x Overall Dia
5.2 Safe Pulling Force with Stocking	N/mm ²	5 x D ²

FRLS properties of Outer Sheath:

1) Oxygen Index: Min 29. 2) Temp. Index: Min 250 Deg C. 3) Halogen acid gas generation: Max.20% by weight. 4) Smoke density rating: 60% Max.

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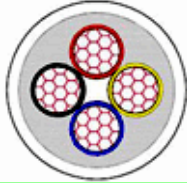
DESCRIPTION	Unit	4C X 6mm ² , 2XY-FRLSH
1 Cable Details		
1.1 kV, Stranded circular, Annealed copper conductor, XLPE insulated, Extruded PVC inner sheathed, Overall PVC sheathed - FRLSH cable, Confirming to IS 7098-1.		
2 Material & Dimensional Details		
2.1 Number of Cores	No.	4
2.2 Conductor Cross-Sectional Area	mm ²	6
2.3 Conductor :		
2.3.1 Material		Annealed copper conductor
2.3.2 Conductor Cross-Sectional Area (Ph/N)	mm ²	6
2.3.3 Conductor Flexibility class		Class-2
2.3.4 Shape		Stranded circular
2.4 Insulation :		
2.4.1 Material		XLPE
2.4.2 Average Thickness	mm	0.70
2.4.3 Minimum Thickness	mm	0.53
2.4.4 Core Identification		As per IS 7098-1
2.5 Inner Sheath :		
2.5.1 Material		Extruded PVC ST 2
2.5.2 Minimum Thickness	mm	0.3
2.6 Armour :		
2.6.1 Material		No Armour
2.6.2 Nominal Size of Armour	mm	Not applicable
2.7 Outer Sheath:		
2.7.1 Material		FRLS
2.7.2 Nominal Thickness	mm	1.80
2.7.3 Colour		Black
2.7.4 Approximate Overall Diameter	mm	16.0 ± 2.0
3 Electrical Parameters		
3.1 Rated Voltage	kV	1.1
3.2 Maximum Conductor DC Resistance at 20°C	ohm/km	3.0800
3.3 Permissible Current Rating in Air at 40°C	A	52
3.4 Maximum Continuous Operating Temperature	°C	90
3.5 Maximum Short Circuit Temperature	°C	250
3.6 Short Circuit Current Rating of Conductor for 1sec.	kA	0.86
4 Packaging		
4.1 Standard Delivery Length	metres	1000
4.2 Individual & overall quantity tolerance	%	± 5
5 Installation Guidelines		
5.1 Minimum Bending Radius	mm	12 x Overall Dia
5.2 Safe Pulling Force with Stocking	N/mm ²	5 x D ²

FRLS properties of Outer Sheath:

1) Oxygen Index: Min 29. 2) Temp. Index: Min 250 Deg C. 3) Halogen acid gas generation: Max.20% by weight. 4) Smoke density rating: 60% Max.

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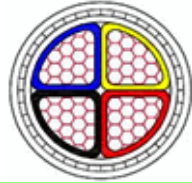
DESCRIPTION	Unit	4C X 10mm ² , 2XY-FRLSH
1 Cable Details		
1.1 kV, Stranded circular, Annealed copper conductor, XLPE insulated, Extruded PVC inner sheathed, Overall PVC sheathed - FRLSH cable, Confirming to IS 7098-1.		
2 Material & Dimensional Details		
2.1 Number of Cores	No.	4
2.2 Conductor Cross-Sectional Area	mm ²	10
2.3 Conductor :		
2.3.1 Material		Annealed copper conductor
2.3.2 Conductor Cross-Sectional Area (Ph/N)	mm ²	10
2.3.3 Conductor Flexibility class		Class-2
2.3.4 Shape		Stranded circular
2.4 Insulation :		
2.4.1 Material		XLPE
2.4.2 Average Thickness	mm	0.70
2.4.3 Minimum Thickness	mm	0.53
2.4.4 Core Identification		As per IS 7098-1
2.5 Inner Sheath :		
2.5.1 Material		Extruded PVC ST 2
2.5.2 Minimum Thickness	mm	0.3
2.6 Armour :		
2.6.1 Material		No Armour
2.6.2 Nominal Size of Armour	mm	Not applicable
2.7 Outer Sheath:		
2.7.1 Material		FRLS
2.7.2 Nominal Thickness	mm	1.80
2.7.3 Colour		Black
2.7.4 Approximate Overall Diameter	mm	19.0 ± 2.0
3 Electrical Parameters		
3.1 Rated Voltage	kV	1.1
3.2 Maximum Conductor DC Resistance at 20°C	ohm/km	1.8300
3.3 Permissible Current Rating in Air at 40°C	A	70
3.4 Maximum Continuous Operating Temperature	°C	90
3.5 Maximum Short Circuit Temperature	°C	250
3.6 Short Circuit Current Rating of Conductor for 1sec.	kA	1.43
4 Packaging		
4.1 Standard Delivery Length	metres	1000
4.2 Individual & overall quantity tolerance	%	± 5
5 Installation Guidelines		
5.1 Minimum Bending Radius	mm	12 x Overall Dia
5.2 Safe Pulling Force with Stocking	N/mm ²	5 x D ²

FRLS properties of Outer Sheath:

1) Oxygen Index: Min 29. 2) Temp. Index: Min 250 Deg C. 3) Halogen acid gas generation: Max.20% by weight. 4) Smoke density rating: 60% Max.

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Property of Ravin Cables Ltd

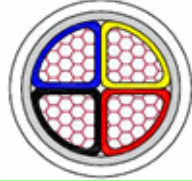
DESCRIPTION	Unit	4C X 16mm ² , 2XFY-FRLSH
1 Cable Details		
1.1 kV, Stranded sector shaped, Annealed copper conductor, XLPE insulated, Extruded PVC inner sheathed, Galvanised steel flat strip armour, Overall PVC sheathed - FRLSH cable, Confirming to IS 7098-1.		
2 Material & Dimensional Details		
2.1 Number of Cores	No.	4
2.2 Conductor Cross-Sectional Area	mm ²	16
2.3 Conductor :		
2.3.1 Material		Annealed copper conductor
2.3.2 Conductor Cross-Sectional Area (Ph/N)	mm ²	16
2.3.3 Conductor Flexibility class		Class-2
2.3.4 Shape		Stranded compact sector
2.4 Insulation :		
2.4.1 Material		XLPE
2.4.2 Average Thickness	mm	0.70
2.4.3 Minimum Thickness	mm	0.53
2.4.4 Core Identification		As per IS 7098-1
2.5 Inner Sheath :		
2.5.1 Material		Extruded PVC ST 2
2.5.2 Minimum Thickness	mm	0.3
2.6 Armour :		
2.6.1 Material		Galvanised steel flat Strip
2.6.2 Nominal Size of Armour	mm	4.0 X 0.8
2.7 Outer Sheath:		
2.7.1 Material		FRLS
2.7.2 Minimum Thickness	mm	1.40
2.7.3 Colour		Black
2.7.4 Approximate Overall Diameter	mm	21.0 ± 2.0
3 Electrical Parameters		
3.1 Rated Voltage	kV	1.1
3.2 Maximum Conductor DC Resistance at 20°C	ohm/km	1.1500
3.3 Permissible Current Rating in Air at 40°C	A	89
3.4 Maximum Continuous Operating Temperature	°C	90
3.5 Maximum Short Circuit Temperature	°C	250
3.6 Short Circuit Current Rating of Conductor for 1sec.	kA	2.29
4 Packaging		
4.1 Standard Delivery Length	metres	1000
4.2 Individual & overall quantity tolerance	%	± 5
5 Installation Guidelines		
5.1 Minimum Bending Radius	mm	12 x Overall Dia
5.2 Safe Pulling Force with Stocking	N/mm ²	9 x D ²

FRLS properties of Outer Sheath:

1) Oxygen Index: Min 29. 2) Temp. Index: Min 250 Deg C. 3) Halogen acid gas generation: Max.20% by weight. 4) Smoke density rating: 60% Max.

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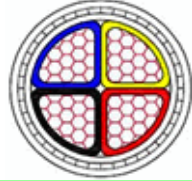
DESCRIPTION	Unit	4C X 16mm ² , 2XY-FRLSH
1 Cable Details		
1.1 kV, Stranded sector shaped, Annealed copper conductor, XLPE insulated, Extruded PVC inner sheathed, Overall PVC sheathed - FRLSH cable, Confirming to IS 7098-1.		
2 Material & Dimensional Details		
2.1 Number of Cores	No.	4
2.2 Conductor Cross-Sectional Area	mm ²	16
2.3 Conductor :		
2.3.1 Material		Annealed copper conductor
2.3.2 Conductor Cross-Sectional Area (Ph/N)	mm ²	16
2.3.3 Conductor Flexibility class		Class-2
2.3.4 Shape		Stranded compact sector
2.4 Insulation :		
2.4.1 Material		XLPE
2.4.2 Average Thickness	mm	0.70
2.4.3 Minimum Thickness	mm	0.53
2.4.4 Core Identification		As per IS 7098-1
2.5 Inner Sheath :		
2.5.1 Material		Extruded PVC
2.5.2 Minimum Thickness	mm	0.3
2.6 Armour :		
2.6.1 Material		No Armour
2.6.2 Nominal Size of Armour	mm	Not applicable
2.7 Outer Sheath:		
2.7.1 Material		FRLS
2.7.2 Nominal Thickness	mm	1.80
2.7.3 Colour		Black
2.7.4 Approximate Overall Diameter	mm	19.0 ± 2.0
3 Electrical Parameters		
3.1 Rated Voltage	kV	1.1
3.2 Maximum Conductor DC Resistance at 20°C	ohm/km	1.1500
3.3 Permissible Current Rating in Air at 40°C	A	89
3.4 Maximum Continuous Operating Temperature	°C	90
3.5 Maximum Short Circuit Temperature	°C	250
3.6 Short Circuit Current Rating of Conductor for 1sec.	kA	2.29
4 Packaging		
4.1 Standard Delivery Length	metres	1000
4.2 Individual & overall quantity tolerance	%	± 5
5 Installation Guidelines		
5.1 Minimum Bending Radius	mm	12 x Overall Dia
5.2 Safe Pulling Force with Stocking	N/mm ²	5 x D ²

FRLS properties of Outer Sheath:

1) Oxygen Index: Min 29. 2) Temp. Index: Min 250 Deg C. 3) Halogen acid gas generation: Max.20% by weight. 4) Smoke density rating: 60% Max.

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Property of Ravin Cables Ltd

DESCRIPTION	Unit	4C X 25mm ² , 2XFY-FRLSH
1 Cable Details		
1.1 kV, Stranded sector shaped, Annealed copper conductor, XLPE insulated, Extruded PVC inner sheathed, Galvanised steel flat strip armour, Overall PVC sheathed - FRLSH cable, Confirming to IS 7098-1.		
2 Material & Dimensional Details		
2.1 Number of Cores	No.	4
2.2 Conductor Cross-Sectional Area	mm ²	25
2.3 Conductor :		
2.3.1 Material		Annealed copper conductor
2.3.2 Conductor Cross-Sectional Area (Ph/N)	mm ²	25
2.3.3 Conductor Flexibility class		Class-2
2.3.4 Shape		Stranded compact sector
2.4 Insulation :		
2.4.1 Material		XLPE
2.4.2 Average Thickness	mm	0.90
2.4.3 Minimum Thickness	mm	0.71
2.4.4 Core Identification		As per IS 7098-1
2.5 Inner Sheath :		
2.5.1 Material		Extruded PVC ST 2
2.5.2 Minimum Thickness	mm	0.3
2.6 Armour :		
2.6.1 Material		Galvanised steel flat Strip
2.6.2 Nominal Size of Armour	mm	4.0 X 0.8
2.7 Outer Sheath:		
2.7.1 Material		FRLS
2.7.2 Minimum Thickness	mm	1.40
2.7.3 Colour		Black
2.7.4 Approximate Overall Diameter	mm	24.0 ± 2.0
3 Electrical Parameters		
3.1 Rated Voltage	kV	1.1
3.2 Maximum Conductor DC Resistance at 20°C	ohm/km	0.7270
3.3 Permissible Current Rating in Air at 40°C	A	119
3.4 Maximum Continuous Operating Temperature	°C	90
3.5 Maximum Short Circuit Temperature	°C	250
3.6 Short Circuit Current Rating of Conductor for 1sec.	kA	3.58
4 Packaging		
4.1 Standard Delivery Length	metres	1000
4.2 Individual & overall quantity tolerance	%	± 5
5 Installation Guidelines		
5.1 Minimum Bending Radius	mm	12 x Overall Dia
5.2 Safe Pulling Force with Stocking	N/mm ²	9 x D ²

FRLS properties of Outer Sheath:

1) Oxygen Index: Min 29. 2) Temp. Index: Min 250 Deg C. 3) Halogen acid gas generation: Max.20% by weight. 4) Smoke density rating: 60% Max.

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Property of Ravin Cables Ltd

Customer: Antelec Limited
Offer ref.No.: QL-476
Project: Hansalpur, Gujarat Line - D for Suzuki Motc

DESCRIPTION	Unit	4C X 25mm ² , 2XY-FRLSH
1 Cable Details		
1.1 kV, Stranded sector shaped, Annealed copper conductor, XLPE insulated, Extruded PVC inner sheathed, Overall PVC sheathed - FRLSH cable, Confirming to IS 7098-1.		
2 Material & Dimensional Details		
2.1 Number of Cores	No.	4
2.2 Conductor Cross-Sectional Area	mm ²	25
2.3 Conductor : IS 8130		
2.3.1 Material		Annealed copper conductor
2.3.2 Conductor Cross-Sectional Area (Ph/N)	mm ²	25
2.3.3 Conductor Flexibility class		Class-2
2.3.4 Shape		Stranded compact sector
2.4 Insulation : IS 7098-1		
2.4.1 Material		XLPE
2.4.2 Average Thickness	mm	0.90
2.4.3 Minimum Thickness	mm	0.71
2.4.4 Core Identification		As per IS 7098-1
2.5 Inner Sheath :		
2.5.1 Material		Extruded PVC
2.5.2 Minimum Thickness	mm	0.3
2.6 Armour : Not Applicable		
2.6.1 Material		No Armour
2.6.2 Nominal Size of Armour	mm	Not applicable
2.7 Outer Sheath: IS 5831		
2.7.1 Material		FRLS
2.7.2 Nominal Thickness	mm	2.00
2.7.3 Colour		Black
2.7.4 Approximate Overall Diameter	mm	23.0 ± 2.0
3 Electrical Parameters		
3.1 Rated Voltage	kV	1.1
3.2 Maximum Conductor DC Resistance at 20°C	ohm/km	0.7270
3.3 Permissible Current Rating in Air at 40°C	A	119
3.4 Maximum Continuous Operating Temperature	°C	90
3.5 Maximum Short Circuit Temperature	°C	250
3.6 Short Circuit Current Rating of Conductor for 1sec.	kA	3.58
4 Packaging		
4.1 Standard Delivery Length	metres	1000
4.2 Individual & overall quantity tolerance	%	± 5
5 Installation Guidelines IS 1255		
5.1 Minimum Bending Radius	mm	12 x Overall Dia
5.2 Safe Pulling Force with Stocking	N/mm ²	5 x D ²

FRLS properties of Outer Sheath:

1) Oxygen Index: Min 29. 2) Temp. Index: Min 250 Deg C. 3) Halogen acid gas generation: Max.20% by weight. 4) Smoke density rating: 60% Max.

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Property of Ravin Cables Ltd

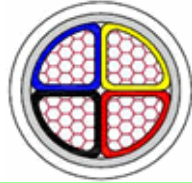
DESCRIPTION	Unit	4C X 35mm ² , 2XY-FRLSH
1 Cable Details		
1.1 kV, Stranded sector shaped, Annealed copper conductor, XLPE insulated, Extruded PVC inner sheathed, Overall PVC sheathed - FRLSH cable, Confirming to IS 7098-1.		
2 Material & Dimensional Details		
2.1 Number of Cores	No.	4
2.2 Conductor Cross-Sectional Area	mm ²	35
2.3 Conductor :		
2.3.1 Material		Annealed copper conductor
2.3.2 Conductor Cross-Sectional Area (Ph/N)	mm ²	35
2.3.3 Conductor Flexibility class		Class-2
2.3.4 Shape		Stranded compact sector
2.4 Insulation :		
2.4.1 Material		XLPE
2.4.2 Average Thickness	mm	0.90
2.4.3 Minimum Thickness	mm	0.71
2.4.4 Core Identification		As per IS 7098-1
2.5 Inner Sheath :		
2.5.1 Material		Extruded PVC
2.5.2 Minimum Thickness	mm	0.3
2.6 Armour :		
2.6.1 Material		No Armour
2.6.2 Nominal Size of Armour	mm	Not applicable
2.7 Outer Sheath:		
2.7.1 Material		FRLS
2.7.2 Nominal Thickness	mm	2.00
2.7.3 Colour		Black
2.7.4 Approximate Overall Diameter	mm	25.0 ± 2.0
3 Electrical Parameters		
3.1 Rated Voltage	kV	1.1
3.2 Maximum Conductor DC Resistance at 20°C	ohm/km	0.5240
3.3 Permissible Current Rating in Air at 40°C	A	147
3.4 Maximum Continuous Operating Temperature	°C	90
3.5 Maximum Short Circuit Temperature	°C	250
3.6 Short Circuit Current Rating of Conductor for 1sec.	kA	5.00
4 Packaging		
4.1 Standard Delivery Length	metres	1000
4.2 Individual & overall quantity tolerance	%	± 5
5 Installation Guidelines		
5.1 Minimum Bending Radius	mm	12 x Overall Dia
5.2 Safe Pulling Force with Stocking	N/mm ²	5 x D ²

FRLS properties of Outer Sheath:

1) Oxygen Index: Min 29. 2) Temp. Index: Min 250 Deg C. 3) Halogen acid gas generation: Max.20% by weight. 4) Smoke density rating: 60% Max.

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Property of Ravin Cables Ltd

DESCRIPTION	Unit	4C X 50mm ² , 2XY-FRLSH
1 Cable Details		
1.1 kV, Stranded sector shaped, Annealed copper conductor, XLPE insulated, Extruded PVC inner sheathed, Overall PVC sheathed - FRLSH cable, Confirming to IS 7098-1.		
2 Material & Dimensional Details		
2.1 Number of Cores	No.	4
2.2 Conductor Cross-Sectional Area	mm ²	50
2.3 Conductor :		
2.3.1 Material		Annealed copper conductor
2.3.2 Conductor Cross-Sectional Area (Ph/N)	mm ²	50
2.3.3 Conductor Flexibility class		Class-2
2.3.4 Shape		Stranded compact sector
2.4 Insulation :		
2.4.1 Material		XLPE
2.4.2 Average Thickness	mm	1.00
2.4.3 Minimum Thickness	mm	0.8
2.4.4 Core Identification		As per IS 7098-1
2.5 Inner Sheath :		
2.5.1 Material		Extruded PVC
2.5.2 Minimum Thickness	mm	0.3
2.6 Armour :		
2.6.1 Material		No Armour
2.6.2 Nominal Size of Armour	mm	Not applicable
2.7 Outer Sheath:		
2.7.1 Material		FRLS
2.7.2 Nominal Thickness	mm	2.00
2.7.3 Colour		Black
2.7.4 Approximate Overall Diameter	mm	28.0 ± 2.0
3 Electrical Parameters		
3.1 Rated Voltage	kV	1.1
3.2 Maximum Conductor DC Resistance at 20°C	ohm/km	0.3870
3.3 Permissible Current Rating in Air at 40°C	A	179
3.4 Maximum Continuous Operating Temperature	°C	90
3.5 Maximum Short Circuit Temperature	°C	250
3.6 Short Circuit Current Rating of Conductor for 1sec.	kA	7.15
4 Packaging		
4.1 Standard Delivery Length	metres	1000
4.2 Individual & overall quantity tolerance	%	± 5
5 Installation Guidelines		
5.1 Minimum Bending Radius	mm	12 x Overall Dia
5.2 Safe Pulling Force with Stocking	N/mm ²	5 x D ²

FRLS properties of Outer Sheath:

1) Oxygen Index: Min 29. 2) Temp. Index: Min 250 Deg C. 3) Halogen acid gas generation: Max.20% by weight. 4) Smoke density rating: 60% Max.

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Property of Ravin Cables Ltd

DESCRIPTION	Unit	19C X 2.5mm ² , 2XFY-FRLSH
1 Cable Details		
1.1 kV, Stranded circular, Annealed copper conductor, XLPE insulated, Extruded PVC inner sheathed, Galvanised steel flat strip armour, Overall PVC sheathed - FRLSH cable, Confirming to IS 7098-1.		
2 Material & Dimensional Details		
2.1 Number of Cores	No.	19
2.2 Conductor Cross-Sectional Area	mm ²	2.5
2.3 Conductor :		
2.3.1 Material		Annealed copper conductor
2.3.2 Conductor Cross-Sectional Area (Ph/N)	mm ²	2.5
2.3.3 Conductor Flexibility class		Class-2
2.3.4 Shape		Stranded circular
2.4 Insulation :		
2.4.1 Material		XLPE
2.4.2 Average Thickness	mm	0.70
2.4.3 Minimum Thickness	mm	0.53
2.4.4 Core Identification		As per IS 7098-1
2.5 Inner Sheath :		
2.5.1 Material		Extruded PVC
2.5.2 Minimum Thickness	mm	0.3
2.6 Armour :		
2.6.1 Material		Galvanised steel flat Strip
2.6.2 Nominal Size of Armour	mm	4.0 X 0.8
2.7 Outer Sheath:		
2.7.1 Material		FRLS
2.7.2 Minimum Thickness	mm	1.40
2.7.3 Colour		Black
2.7.4 Approximate Overall Diameter	mm	24.0 ± 2.0
3 Electrical Parameters		
3.1 Rated Voltage	kV	1.1
3.2 Maximum Conductor DC Resistance at 20°C	ohm/km	7.4100
3.3 Permissible Current Rating in Air at 40°C	A	16
3.4 Maximum Continuous Operating Temperature	°C	90
3.5 Maximum Short Circuit Temperature	°C	250
3.6 Short Circuit Current Rating of Conductor for 1sec.	kA	0.36
4 Packaging		
4.1 Standard Delivery Length	metres	1000
4.2 Individual & overall quantity tolerance	%	± 5
5 Installation Guidelines		
5.1 Minimum Bending Radius	mm	12 x Overall Dia
5.2 Safe Pulling Force with Stocking	N/mm ²	9 x D ²

FRLS properties of Outer Sheath:

1) Oxygen Index: Min 29. 2) Temp. Index: Min 250 Deg C. 3) Halogen acid gas generation: Max.20% by weight. 4) Smoke density rating: 60% Max.

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Property of Ravin Cables Ltd

DESCRIPTION	Unit	3.5C x 400 mm ² : A2XY/FR-LSH
1 Cross Sectional Drawing		Description
		Stranded Sector Shaped H4 Grade Aluminium Conductor, XLPE Insulated, Cores Laid-up together, Extruded PVC Inner Sheath, Un-Armoured, Overall PVC ST 2 FR-LSH Outer Sheathed 1.1 kV LT Cables Confirming to IS 7098-1.
2 Material & Dimensional Details		
2.1 Number of Cores		3.5
2.2 Cross-Sectional Area	mm ²	400
2.3 Conductor :		IS 8130
2.3.1 Material		H4 Grade Aluminium
2.3.2 Conductor Cross-Sectional Area (Ph/N)	mm ²	400 / 185
2.3.3 Conductor Flexibility class		Class - 2
2.3.4 Shape	No/mm	Sector Shaped
2.4 Insulation :		IS 7098-1
2.4.1 Material		XLPE
2.4.2 Nominal Thickness (Ph/N)	mm	2 / 1.6
2.4.3 Minimum Thickness (Ph/N)	mm	1.7 / 1.34
2.4.4 Core Identification		As per IS 7098-1
2.5 Inner Sheath :		
2.5.1 Material		Extruded PVC
2.5.2 Minimum Thickness	mm	0.7
2.6 Armour :		
2.6.1 Material		Un-Armoured
2.6.2 Nominal Size of Armour	mm	Not Applicable
2.7 Outer Sheath :		IS 5831
2.7.1 Material		PVC ST 2 FR-LSH
2.7.2 Nominal Thickness	mm	3.40
2.7.3 Colour		BLACK
2.7.4 Approximate Overall Diameter	mm	65.0 ± 3
3 Electrical Parameters		
3.1 Rated Voltage	kV	1.1
3.2 Maximum Conductor DC Resistance at 20°C	Ω/km	0.0778
3.3 Permissible Current Rating in Air at 40°C	A	526
3.4 Maximum Continuous Operating Temperature	°C	90
3.5 Maximum Short Circuit Temperature	°C	250
3.6 Short Circuit Current Rating of Conductor for 1sec.	kA	37.60
4 Packaging		
4.1 Standard Delivery Length	m	500
4.2 Individual & Overall Quantity Tolerance	%	± 5
5 Installation Guidelines		IS 1255
5.1 Minimum Bending Radius	mm	12 x Overall Dia.
5.2 Safe Pulling Force with Stocking	Newtons	5 x D ²

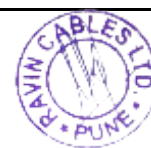
Note: Given weights & dimensions are for guidance only,subject to revision without prior intimation.

Marking: Logo Ravin, Electric, Voltage Grade, ISI, IS 7098-1, CM/L No, FRLS, Year with Sequential marking

FR-LSH Properties of Outer sheath

1) Oxygen Index: Min. 29; 2) Halogen acid gas generation: Max. 20% by weight

3) Temp. Index: Min. 250 °C; 4) Smoke density rating: 60% Max.



DESCRIPTION		Unit	3.5C x 50 mm ² : 2XY/FR-LSH
1 Cross Sectional Drawing			Description
			Stranded Sector Shaped Plain Annealed Copper Conductor, XLPE Insulated, Cores Laid-up together, Extruded PVC Inner Sheath, Un-Armoured, Overall PVC ST 2 FR-LSH Outer Sheathed 1.1 kV LT Cables Confirming to IS 7098-1.
2 Material & Dimensional Details			
2.1 Number of Cores			3.5
2.2 Cross-Sectional Area	mm ²		50
2.3 Conductor :			IS 8130
2.3.1 Material			Plain Annealed Copper
2.3.2 Conductor Cross-Sectional Area (Ph/N)	mm ²		50 / 25
2.3.3 Conductor Flexibility class			Class - 2
2.3.4 Shape	No/mm		Sector Shaped
2.4 Insulation :			IS 7098-1
2.4.1 Material			XLPE
2.4.2 Nominal Thickness (Ph/N)	mm		1 / 0.9
2.4.3 Minimum Thickness (Ph/N)	mm		0.8 / 0.71
2.4.4 Core Identification			As per IS 7098-1
2.5 Inner Sheath :			
2.5.1 Material			Extruded PVC
2.5.2 Minimum Thickness	mm		0.3
2.6 Armour :			
2.6.1 Material			Un-Armoured
2.6.2 Nominal Size of Armour	mm		Not Applicable
2.7 Outer Sheath :			IS 5831
2.7.1 Material			PVC ST 2 FR-LSH
2.7.2 Nominal Thickness	mm		2.00
2.7.3 Colour			BLACK
2.7.4 Approximate Overall Diameter	mm		26.0 ± 2
3 Electrical Parameters			
3.1 Rated Voltage	kV		1.1
3.2 Maximum Conductor DC Resistance at 20°C	Ω/km		0.387
3.3 Permissible Current Rating in Air at 40°C	A		179
3.4 Maximum Continuous Operating Temperature	°C		90
3.5 Maximum Short Circuit Temperature	°C		250
3.6 Short Circuit Current Rating of Conductor for 1sec.	kA		7.15
4 Packaging			
4.1 Standard Delivery Length	m		1000
4.2 Individual & Overall Quantity Tolerance	%		± 5
5 Installation Guidelines			IS 1255
5.1 Minimum Bending Radius	mm		12 x Overall Dia.
5.2 Safe Pulling Force with Stocking	Newtons		5 x D ²

Note: Given weights & dimensions are for guidance only,subject to revision without prior intimation.

Marking: Logo Ravin, Electric, Voltage Grade, ISI, IS 7098-1, CM/L No, FRLS, Year with Sequential marking

FR-LSH Properties of Outer sheath

1) Oxygen Index: Min. 29; 2) Halogen acid gas generation: Max. 20% by weight

3) Temp. Index: Min. 250 °C; 4) Smoke density rating: 60% Max.



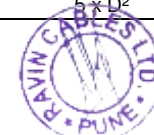
DESCRIPTION	Unit	1C x 2.5 mm ² : Y/FR-LSH	1C x 6 mm ² : Y/FR-LSH	1C x 16 mm ² : Y/FR-LSH	1C x 25 mm ² : Y/FR-LSH	1C x 50 mm ² : Y/FR-LSH
1 Cross Sectional Drawing		Description				
		Stranded Flexible Circular Plain Annealed Copper Conductor, FRLSH PVC Type D Insulation, unsheathed 1100 V LT Cables Confirming to IS 694.				
2 Construction Details (Material)						
2.1 Conductor		Plain Annealed Copper	Plain Annealed Copper	Plain Annealed Copper	Plain Annealed Copper	Plain Annealed Copper
2.2 Insulation		FRLS PVC Type D	FRLS PVC Type D	FRLS PVC Type D	FRLS PVC Type D	FRLS PVC Type D
3 Dimensional Details						
3.1 Number of Cores		1	1	1	1	1
3.2 Conductor Cross-Sectional Area	mm ²	2.5	6	16	25	50
3.3 Approximate Overall Diameter	mm	4	5.5	7.5	10	12.5
3.5 Conductor :						
3.5.1 Conductor Flexibility class		Class-5	Class-5	Class-5	Class-5	Class-5
3.5.2 Max. wire size	mm	0.26	0.31	0.41	0.41	0.41
3.5.3 Shape		Flexible Circular	Flexible Circular	Flexible Circular	Flexible Circular	Flexible Circular
3.6 Insulation :						
3.6.1 Average Thickness	mm	0.7	0.8	1	1.2	1.4
3.6.2 Minimum Thickness	mm	0.53	0.62	0.8	0.98	1.16
3.6.3 Core Identification		Green	Green	Green	Green	Green
4 Electrical Parameters						
4.1 Rated Voltage	V	1100	1100	1100	1100	1100
4.2 Maximum Conductor DC Resistance at 20°C	Ω/km	7.98	3.3	1.21	0.78	0.386
4.3 Permissible Current Rating in Air at 40°C for 1-Ph	A	22	37	68	86	145
4.5 Maximum Continuous Operating Temperature	°C	70	70	70	70	70
4.6 Maximum Short Circuit Temperature	°C	160	160	160	160	160
4.7 Short Circuit Current Rating of Conductor for 1sec.	kA	0.29	0.69	1.84	2.88	5.75
5 Packaging						
5.1 Standard Delivery Length	m	1000	1000	1000	1000	1000
5.2 Individual & overall quantity tolerance	%	± 5	± 5	± 5	± 5	± 5
6 Applicable Standards						
6.1 Cable		IS 694	IS 694	IS 694	IS 694	IS 694
6.2 Conductor		IS 8130	IS 8130	IS 8130	IS 8130	IS 8130
6.3 Insulation		IS 5831	IS 5831	IS 5831	IS 5831	IS 5831
7 Installation Guidelines						
7.1 Minimum Bending Radius	mm	8 x Overall Dia.	8 x Overall Dia.	8 x Overall Dia.	8 x Overall Dia.	8 x Overall Dia.
7.2 Safe Pulling Force with Stocking	Newtons	5 x D ²	5 x D ²	5 x D ²	5 x D ²	5 x D ²

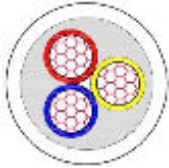
Note: Given weights & dimensions are for guidance only, subject to revision without prior intimation.

FR-LSH Properties of Outer sheath

1) Oxygen Index: Min. 29; 2) Halogen acid gas generation: Max. 20% by weight

3) Temp. Index: Min. 250 °C; 4) Smoke density rating: 60% Max.



DESCRIPTION	Unit	3C x 1.5 mm ² : YY/FR-LSH	3C x 2.5 mm ² : YY/FR-LSH
1 Cross Sectional Drawing		Description	
		Flexible Circular Annealed Plain Copper Conductor, PVC Type D Insulation, Cores Laid-up together, Overall FRLSH PVC ST3 Sheathed 1100 V LT Cables Generally Conforming to IS 694.	
2 Construction Details (Material)			
2.1 Conductor		Plain Annealed Copper	Plain Annealed Copper
2.2 Insulation		PVC Type D	PVC Type D
2.5 Sheath		FRLSH PVC ST 3	FRLSH PVC ST 3
3 Dimensional Details			
3.1 Number of Cores		3	3
3.2 Conductor Cross-Sectional Area	mm ²	1.5	2.5
3.3 Approximate Overall Diameter	mm	9.0	10.5
3.5 Conductor :			
3.5.1 Conductor Flexibility class		Flexible	Flexible
3.5.2 Max. wire size	No/mm	0.26	0.26
3.5.3 Shape		Circular	Circular
3.6 Insulation :			
3.6.1 Average Thickness	mm	0.6	0.7
3.6.2 Minimum Thickness	mm	0.44	0.53
3.6.3 Core Identification		Red, Yellow, Blue	Red, Yellow, Blue
3.7 Outer Sheath :			
3.7.1 Nominal Thickness	mm	0.9	1
3.7.2 Colour		Black	Black
4 Electrical Parameters			
4.1 Rated Voltage Uo/U	V	1100	1100
4.2 Maximum Conductor DC Resistance at 20°C	Ω/km	13.3	7.98
4.3 Permissible Current Rating in Air at 40°C for 3-Ph	A	14	18
4.5 Maximum Continuous Operating Temperature	°C	70	70
4.6 Maximum Short Circuit Temperature	°C	160	160
4.7 Short Circuit Current Rating of Conductor for 1sec.	kA	0.17	0.29
5 Packaging			
5.1 Standard Delivery Length	m	1000	1000
5.2 Tolerance	%	± 5	± 5
6 Applicable Standards			
6.1 Cable		IS 694	IS 694
6.2 Conductor		IS 8130	IS 8130
6.3 Insulation		IS 5831	IS 5831
6.4 Sheath		IS 5831	IS 5831
7 Installation Guidelines			
7.1 Minimum Bending Radius	mm	8 x Overall Dia.	8 x Overall Dia.
7.2 Safe Pulling Force by Pulling Eye	N/mm ²	50	50

Note: Given weights & dimensions are for guidance only, subject to revision without prior intimation.

FR-LSH Properties of Outer sheath

1) Oxygen Index: Min. 29; 2) Halogen acid gas generation: Max. 20% by weight

3) Temp. Index: Min. 250 °C; 4) Smoke density rating: 60% Max.

